Value-Glamour and Accruals Mispricing:

One Anomaly or Two?

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Agenda

1. Introduction & Paper Recap
2. Signal Replication — Data & Construction
3. Descriptive Statistics & Signal Diagnostics
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Introduction & Paper Recap

Paper: Desai, Rajgopal & Venkatachalam (2004)

Two well-known anomalies in equity markets:

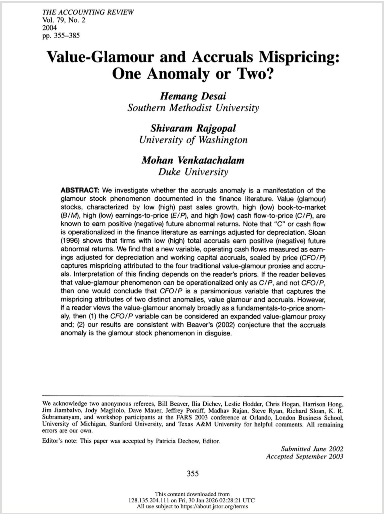
- Value-Glamour: Value stocks (high B/M) outperform glamour stocks (low B/M)
- Accruals: Firms with high accruals earn lower future returns

Key question: Are these one anomaly or two?

Signal: CFO/P (Operating Cash Flow / Price)

- CFO = Earnings + Depreciation − Working-capital accruals
- CFO/P combines fundamentals-to-price ratio with accrual correction
- Higher CFO/P → stronger cash generation → higher future abnormal returns

Finding: CFO/P absorbs both Value–Glamour proxies and Accruals anomaly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Value-Glamour and Accruals Mispricing: One Anomaly or Two?

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ABSTRACT: We investigate whether the accruals anomaly is a manifestation of the glamour stock phenomenon documented in the finance literature. Value (glamour) stocks, characterized by low (high) past sales growth, high (low) book-to-market (*B/M*), high (low) earnings-to-price (*E/P*), and high (low) cash flow-to-price (*C/P*), are known to earn positive (negative) future abnormal returns. Note that "*C*" or cash flow is operationalized in the finance literature as earnings adjusted for depreciation. Sloan (1996) shows that firms with low (high) total accruals earn positive (negative) future abnormal returns. We find that a new variable, operating cash flows measured as earnings adjusted for depreciation and working capital accruals, scaled by price (*CFO/P*) captures mispricing attributed to the four traditional value-glamour proxies and accruals. Interpretation of this finding depends on the reader's priors. If the reader believes that value-glamour phenomenon can be operationalized only as *C/P*, and not *CFO/P*, then one would conclude that *CFO/P* is a parsimonious variable that captures the mispricing attributes of two distinct anomalies, value glamour and accruals. However, if a reader views the value-glamour anomaly broadly as a fundamentals-to-price anomaly, then (1) the *CFO/P* variable can be considered an expanded value-glamour proxy and; (2) our results are consistent with Beaver's (2002) conjecture that the accruals anomaly is the glamour stock phenomenon in disguise.

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Signal Replication — Data & Construction

Data Sources (WRDS)

CRSP (MSF + MSENAMES)

Monthly returns, price, shares outstanding

Compustat (FUNDA)

Earnings (IB), depreciation (DP), accrual items

CRSP–Compustat Link (CCM)

GVKEY ↔ PERMNO mapping

Universe & Filters

- Period: 1973–1997 (25 years)
- NYSE, AMEX, NASDAQ
- US common stocks (SHRCD 10, 11)
- Exclude financials (SIC 6000–6999)
- Exclude negative book value
- Sales ≥ $1M

Signal Construction

CFO = Earnings + Depreciation − Working-capital accruals | CFO/P = CFO / Market Equity

Timing: Portfolios formed 4 months after fiscal year-end (avoid look-ahead bias)

Final panel: 87,931 firm-year observations after cleaning

Descriptive Statistics

Panel A: Summary Statistics (after cleaning, N = 87,931)

Signal	Mean	Std Dev	Median	Min	Max
ACC	-0.02	0.12	-0.03	-1.25	1.41
A/P	-0.09	0.53	-0.03	-25.62	11.85
B/M	0.87	0.83	0.67	0.00	24.09
SG	0.25	0.81	0.13	-0.67	37.74
E/P	0.11	0.36	0.11	-16.11	5.38
C/P	0.21	0.36	0.17	-11.93	8.16
CFO/P	**0.20**	**0.50**	**0.15**	**-14.40**	**16.61**
CFO/TA	0.10	0.17	0.12	-2.12	1.43

CFO/P is the primary signal: mean of 0.20 with moderate dispersion (std = 0.50), right-skewed distribution

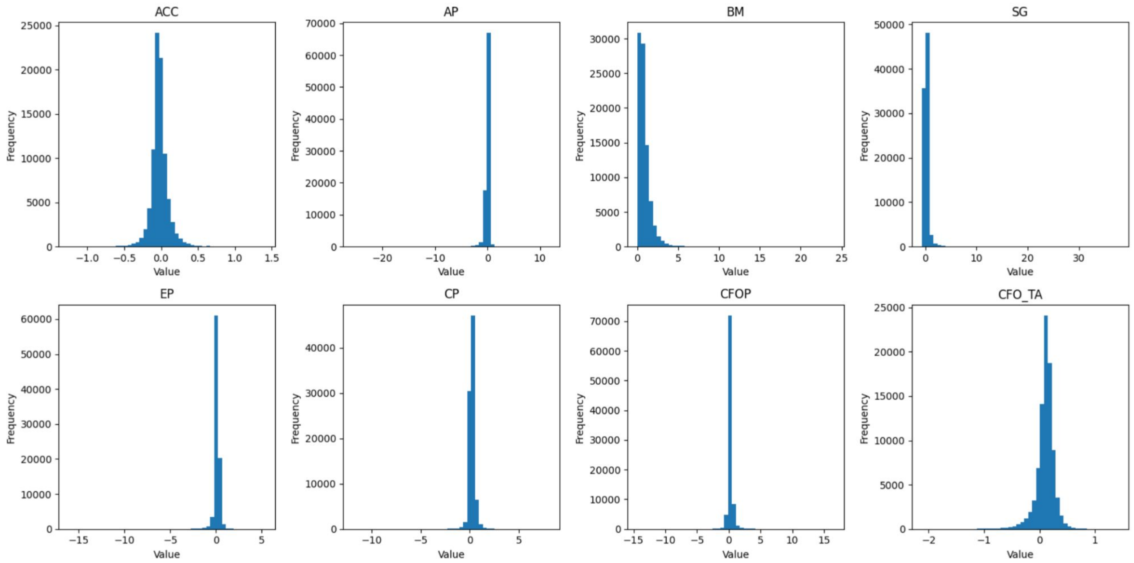
Correlation Matrix

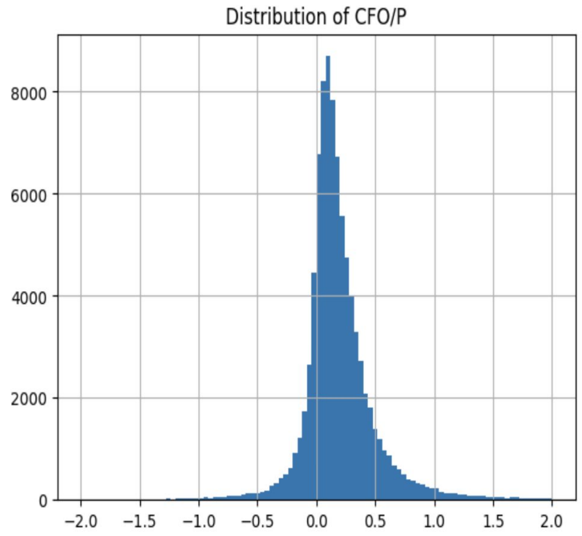
Panel B: Pairwise Correlations (N = 87,931)

	ACC	A/P	B/M	SG	E/P	C/P	CFO/P	CFO/TA
ACC	1.00	**0.50**	-0.08	0.16	0.19	0.07	-0.39	-0.48
A/P	**0.50**	1.00	-0.24	0.07	0.41	0.11	**-0.72**	-0.21
B/M	-0.08	-0.24	1.00	-0.10	0.03	0.28	0.27	-0.04
SG	0.16	0.07	-0.10	1.00	0.01	-0.01	-0.07	-0.09
E/P	0.19	0.41	0.03	0.01	1.00	**0.84**	0.32	0.27
C/P	0.07	0.11	0.28	-0.01	**0.84**	1.00	**0.51**	0.27
CFO/P	-0.39	**-0.72**	0.27	-0.07	0.32	**0.51**	1.00	0.42
CFO/TA	-0.48	-0.21	-0.04	-0.09	0.27	0.27	0.42	1.00

Key: CFO/P strongly negatively correlated with A/P (−0.72) and ACC (−0.39), consistent with the paper’s thesis that CFO/P captures accruals mispricing. High positive correlation with C/P (0.51) confirms overlap with value-glamour proxies.

Signal Distributions

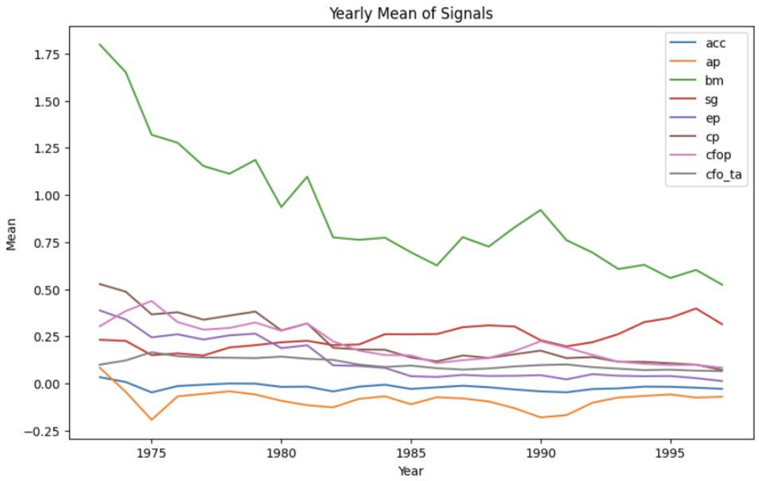




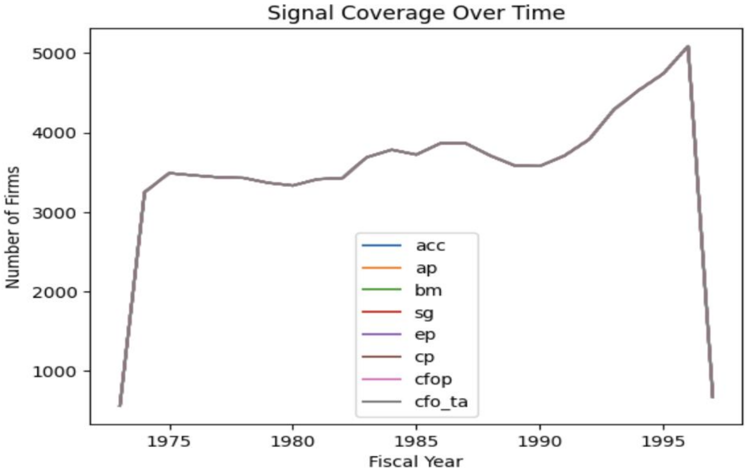
Distributions after winsorization at 0.004%/99.996%. CFO/P shows a roughly symmetric distribution centered near 0.15 with moderate right tail.

Outlier & Stability Analysis

Signal Means Over Time



Signal Coverage Over Time

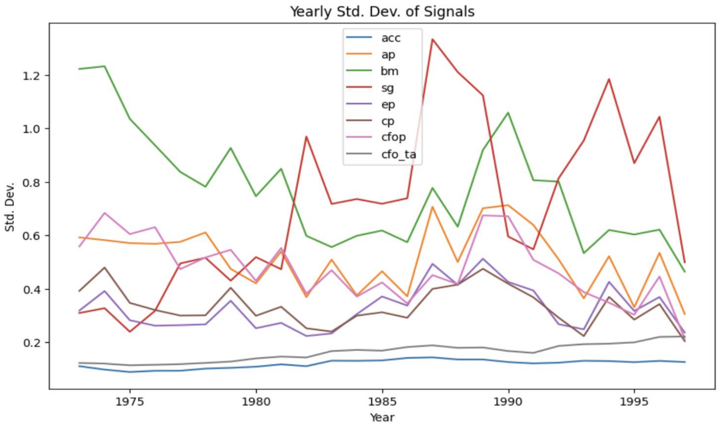


Cleaning steps: Dropped missing values, winsorized extremes at 0.004%/99.996% thresholds, removed sales growth outliers (SG > 48 or SG ≤ −0.7). Final panel: 87,931 firm-year observations. Signal means are stable across time; coverage grows steadily with NASDAQ expansion.

Cross-Sectional Statistics Over Time

Average of yearly cross-sectional statistics (key signals)

Signal	Mean	Std Dev	Median	Min	Max
C/P	0.23	0.33	0.21	-5.13	5.22
CFO/P	**0.21**	**0.47**	**0.17**	**-6.95**	**9.77**
CFO/TA	0.10	0.16	0.12	-1.39	1.04



CFO/P shows stable volatility (~0.3–0.5) across the full sample, making it a reliable sorting variable for portfolio formation.

Comparison with WRDS Prebuilt Signal

Replicated vs. WRDS Prebuilt Signals (N ≈ 86,000 matched firm-months)

Signal	N	Correlation	Mean Abs Diff	Mean Diff
ACC	85,716	0.067	0.073	0.063
B/M	85,832	**0.862**	0.130	0.050
SG	85,532	0.559	0.175	0.035
E/P	86,447	0.696	0.130	0.108
CFO/P	**86,449**	**0.683**	**0.141**	**0.101**
CFO/TA	86,449	**0.847**	0.072	0.050

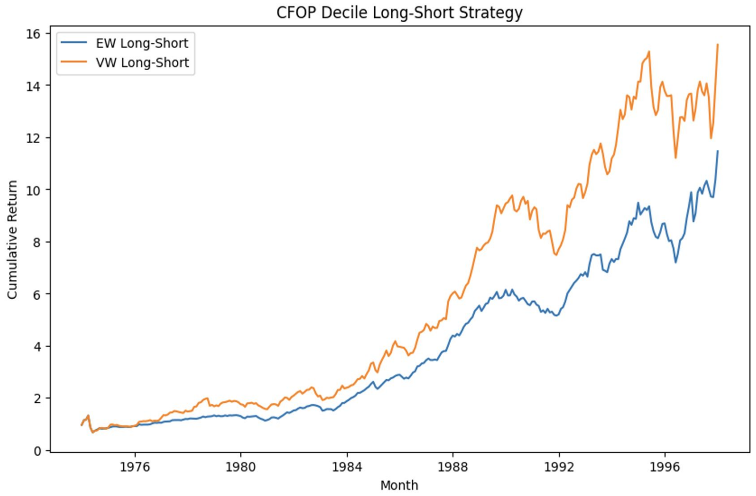
Key Differences

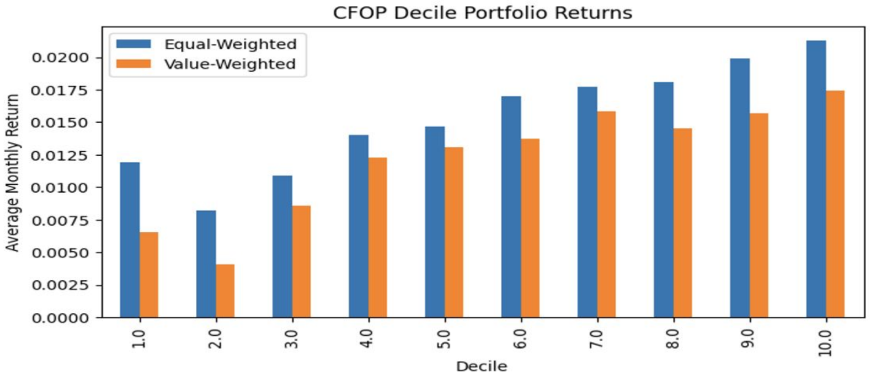
- CFO/TA and B/M show strongest agreement (corr > 0.84), confirming similar construction methodology
- ACC has low correlation (0.067) — WRDS uses a different accruals definition (likely cash-flow statement based vs. our balance-sheet approach)
- CFO/P correlation of 0.683 reflects differences in how working-capital accruals are computed, timing conventions, missing value treatment, winsorization

Portfolio Evaluation — CFO/P Backtest

CFO/P Decile Long-Short Returns (289 months)

	Monthly Return	Mo. Vol	Annual Return	Ann. Vol	t-stat	Sharpe	Hit Rate
EW L/S	0.94%	4.10%	11.3%	14.2%	**3.89**	0.31	66.4%
VW L/S	1.09%	5.04%	13.0%	17.5%	**3.67**	0.35	65.1%





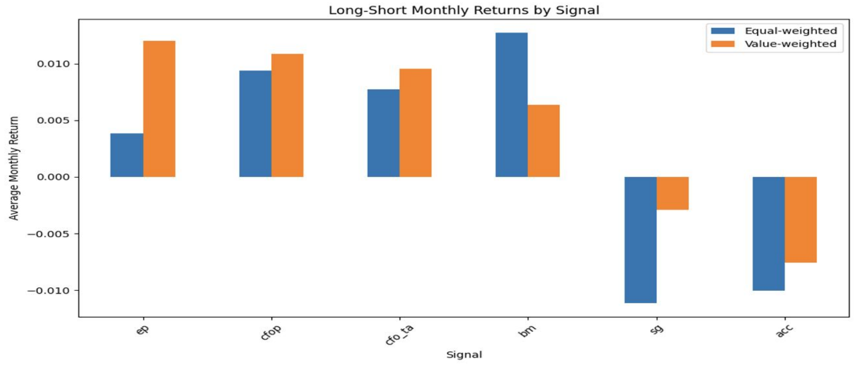
Strong monotonic pattern: D10 (highest CFO/P) returns 2.13% EW / 1.74% VW monthly vs D1 at 1.19% / 0.65%. t-stats > 3.6 confirm high statistical significance.

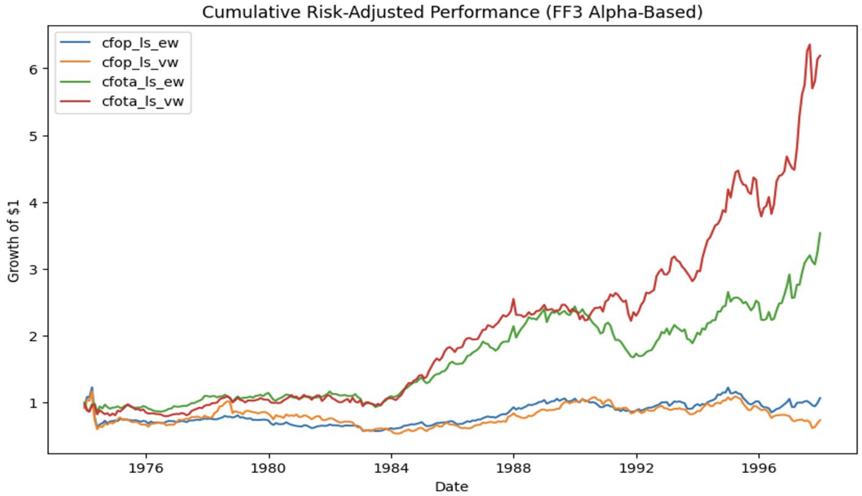
Risk-Adjusted Performance

Fama-French 3-Factor Alpha (Long-Short Portfolios)

$$R_{ls} - Rf = \alpha + \beta_m(MKT\text{-}Rf) + \beta_s(SMB) + \beta_h(HML) + e$$

Strategy	α Monthly	α Annual	t-stat	R^2
CFO/P EW	0.10%	1.3%	0.44	0.11
CFO/P VW	-0.00%	-0.0%	-0.001	0.22
CFO/TA EW	**0.49%**	**5.9%**	**2.45**	0.25
CFO/TA VW	**0.69%**	**8.3%**	**3.21**	0.41





CFO/TA generates significant alpha (VW: 8.3% annual, t=3.21) after Fama-French adjustment. CFO/P raw returns are strong but largely explained by value (HML) exposure. CFO/TA captures profitability beyond standard risk factors.

Risk-Adjusted Performance

Fama-French 3-Factor Alpha (Long-Short Portfolios)

Equal-weight portfolios:

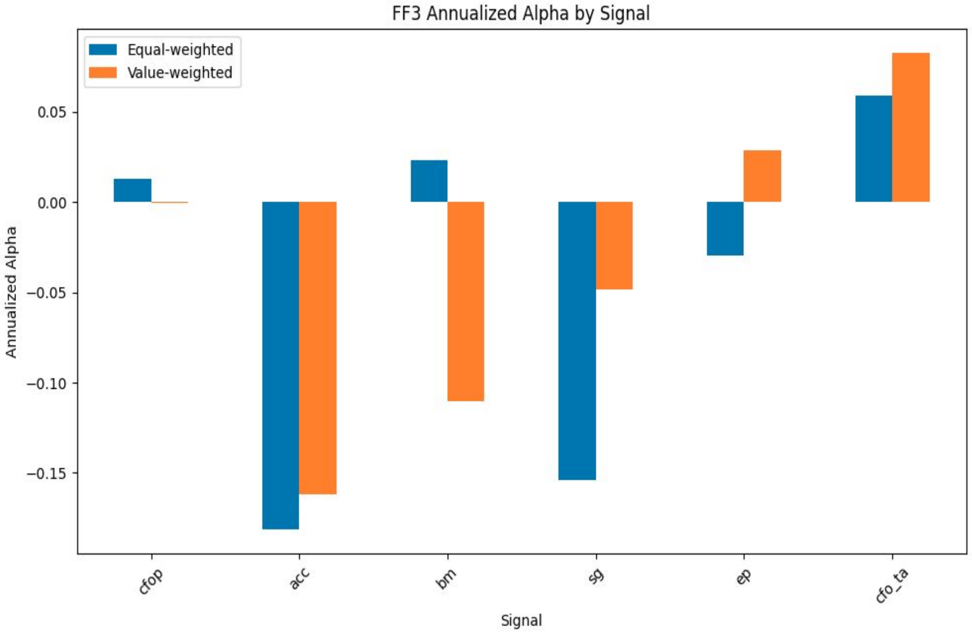
+ Annualized Alpha: CFO/P, BM, CFO/TA

- Annualized Alpha: ACC, SG, E/P

Value-weight portfolios:

+ Annualized Alpha: CFO/TA, E/P

- Annualized Alpha: CFO/P, ACC, SG, BM



Key Takeaways

1. Successful replication

CFO/P and CFO/TA signals replicated from CRSP/Compustat; correlations of 0.68 and 0.85 with WRDS benchmarks

2. CFO/P delivers strong raw returns

Long-short spread of ~0.94–1.09% monthly ($t > 3.6$), monotonic decile pattern, 11–13% annualized

3. CFO/TA generates significant risk-adjusted alpha

VW alpha of 8.3% annual ($t = 3.21$) after FF3 adjustment; CFO/P alpha absorbed by HML exposure

4. Consistent with the paper's thesis

CFO/P unifies value-glamour and accruals anomalies. Cash-flow fundamentals contain persistent pricing information not fully captured by standard risk factors

Appendix

Appendix

Key Risk Adjusted Metrics: CFO/P and CFO/TA LS portfolios

strategy	n_months	mean_monthly	vol_monthly	ann_return	ann_vol	sharpe	t_stat	max_drawdown	best_month	worst_month	hit_rate
cfop_ls_ew	289	0.0094	0.0410	0.1126	0.1419	0.3091	3.8939	-0.4875	0.1866	-0.3492	0.6644
cfop_ls_vw	289	0.0109	0.0504	0.1305	0.1746	0.3538	3.6681	-0.4875	0.1866	-0.3492	0.6505
cfota_ls_ew	289	0.0077	0.0374	0.0928	0.1297	0.1866	3.5117	-0.2908	0.1090	-0.1866	0.6263
cfota_ls_vw	289	0.0095	0.0456	0.1143	0.1578	0.2896	3.5553	-0.3524	0.1561	-0.1866	0.5779

Cumulative risk-adjusted performance per signal

	signal	ew_n_months	ew_mean_monthly_ls	ew_alpha_monthly	ew_alpha_annual	ew_alpha_tstat	ew_r2	vw_n_months	vw_mean_monthly_ls	vw_alpha_monthly	vw_alpha_annual	vw_alpha_tstat	vw_r2
0	cfop	289	0.0094	0.0010	0.0126	0.4367	0.1126	289	0.0108	-0.0000	-0.0005	-0.0152	0.2244
1	acc	289	-0.0101	-0.0151	-0.1813	-7.1600	0.0747	289	-0.0076	-0.0135	-0.1616	-5.1108	0.1087
2	bm	289	0.0127	0.0019	0.0231	1.0407	0.5255	289	0.0063	-0.0092	-0.1104	-4.4093	0.6165
3	sg	289	-0.0111	-0.0128	-0.1538	-5.9687	0.3027	289	-0.0029	-0.0041	-0.0487	-1.5810	0.2902
4	ep	289	0.0038	-0.0025	-0.0297	-0.9696	0.0925	289	0.0120	0.0024	0.0288	0.7359	0.1240
5	cfo_ta	289	0.0077	0.0049	0.0588	2.4464	0.2531	289	0.0095	0.0069	0.0824	3.1837	0.4159